Chapter 11: Infinite Horizon Models

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11.1 Intertemporal Utility Maximization under Poisson Uncertainty

- The chapter starts from optimal consumption-saving under Poisson uncertainty.
- Future consumption is stochastic, so lifetime utility is written in expected present-value form.

$$U(t) = E_t \int_t^{\infty} e^{-\rho(\tau - t)} u(c(\tau))\, d\tau. \tag{11.1.1}$$

- The state variable is household wealth; uncertainty enters through jumps in asset prices.

11.1.1 The setup

- The household budget constraint is derived from the stochastic evolution of wealth.
- Asset returns contain a deterministic component and a Poisson jump component.

$$\mathrm{d}a(t) = \{r(t)a(t) + w(t) - pc(t)\}\mathrm{d}t + \beta a(t)\mathrm{d}q(t). \tag{11.1.2}$$

$$r(t) \equiv \alpha + \frac{\pi(t)}{v(t)}, \qquad \tilde{a}(t) = (1+\beta)a(t). \tag{11.1.3}$$

- The price of consumption is kept constant; wage changes are treated as unanticipated.

11.1.2 Solving by Dynamic Programming: DP1

- Define the value function as the maximum attainable expected utility.
- The continuous-time Bellman equation uses the expected differential of the value function.

$$V(a) \equiv \max_{\{c(\tau)\}} U(t) \quad \text{s.t. (11.1.2)},$$

$$\rho V(a(t)) = \max_{c(t)} \left\{ u(c(t)) + \frac{1}{\mathrm{d}t} E_t \mathrm{d}V(a(t)) \right\}. \tag{11.1.4}$$

11.1.2 Solving by Dynamic Programming: Bellman Equation

- Use the change-of-variable formula for a function of wealth under Poisson jumps.

$$\begin{aligned} \mathrm{d}V(a) &= V'(a)\{ra + w - pc\}\mathrm{d}t + \big[V(a + \beta a) - V(a)\big]\mathrm{d}q, \\ E_t\mathrm{d}V(a(t)) &= V'(a)\{ra + w - pc\}\mathrm{d}t + \big[V(\tilde{a}) - V(a)\big]E_t\mathrm{d}q. \end{aligned}$$

- Since $E_t\mathrm{d}q = \lambda\mathrm{d}t$, the Bellman equation becomes

$$\rho V(a) = \max_{c(t)} \{u(c(t)) + V'(a)[ra + w - pc] + \lambda[V((1+\beta)a) - V(a)]\}. \tag{11.1.5}$$

11.1.2 Solving by Dynamic Programming: FOC

- The first-order condition equates marginal utility of consumption to the shadow value of wealth times the consumption price.

$$u'(c) = V'(a)p. \tag{11.1.6}$$

- This is the usual consumption-saving optimality condition in stochastic continuous time.
- One unit of consumption costs p units of wealth.

11.1.2 Solving by Dynamic Programming: DP2

- Replace controls by their optimal values and differentiate the maximized Bellman equation with respect to wealth.

$$\rho V(a) = u(c(a)) + V'(a)[ra + w - pc(a)] + \lambda[V(\tilde{a}) - V(a)].$$
$$\rho V'(a) = V'(a)r + V''(a)[ra + w - pc] + \lambda\{V'(\tilde{a})[1 + \beta] - V'(a)\}. \qquad (11.1.7)$$

- The jump term requires the derivative of post-jump wealth $\tilde{a} = (1 + \beta)a$.

11.1.2 Solving by Dynamic Programming: Costate Dynamics

- Apply the Poisson change-of-variable formula to the shadow value $V'(a)$.

$$\mathrm{d}V'(a) = V''(a)[ra + w - pc]\mathrm{d}t + \big[V'(\tilde{a}) - V'(a)\big]\mathrm{d}q. \tag{11.1.8}$$

- Substitute the expression for $V''(a)[ra + w - pc]$ from (11.1.7):

$$\begin{aligned}\mathrm{d}V'(a) = &\big\{(\rho - r)V'(a) - \lambda\big[V'(\tilde{a})(1+\beta) - V'(a)\big]\big\}\,\mathrm{d}t \\ &+ \big[V'(\tilde{a}) - V'(a)\big]\mathrm{d}q.\end{aligned}$$

11.1.2 Solving by Dynamic Programming: DP3

- Use the FOC $u'(c) = pV'(a)$ and constant p to express the law of motion in marginal utility units.

$$\begin{aligned} \mathrm{d}u'(c) = &\big\{(\rho - r)u'(c) - \lambda\big[u'(\tilde{c})(1+\beta) - u'(c)\big]\big\}\,\mathrm{d}t \\ &+ \big[u'(\tilde{c}) - u'(c)\big]\mathrm{d}q. \end{aligned} \tag{11.1.9}$$

- The rule contains both a deterministic drift and a stochastic jump component.

11.1.3 The Keynes-Ramsey Rule: Inverse Marginal Utility

- Let $f(\cdot)$ be the inverse of marginal utility: $f(u'(c)) = c$.
- Apply the Poisson CVF to $f(u'(c))$ to obtain consumption dynamics.

$$\begin{aligned}\mathrm{d}f(u'(c)) =& f'(u'(c))\left\{(\rho - r)u'(c) - \lambda[u'(\tilde{c})(1+\beta) - u'(c)]\right\}\mathrm{d}t \\ &+ \{f(u'(\tilde{c})) - f(u'(c))\}\mathrm{d}q,\end{aligned}$$

$$f(u'(\tilde{c})) = \tilde{c}, \qquad f'(u'(c)) = \frac{\mathrm{d}c}{\mathrm{d}u'(c)} = \frac{1}{u''(c)}.$$

11.1.3 The Keynes-Ramsey Rule: General Form

- The optimal consumption rule under Poisson uncertainty is

$$\begin{aligned} \mathrm{d}c =& \frac{1}{u''(c)} \{(\rho - r)u'(c) - \lambda[u'(\tilde{c})(1+\beta) - u'(c)]\} \mathrm{d}t \\ &+ \{\tilde{c} - c\}\mathrm{d}q, \end{aligned}$$

$$\frac{u''(c)}{u'(c)}\mathrm{d}c = \left\{ r - \rho + \lambda \left[\frac{u'(\tilde{c})}{u'(c)}(1+\beta) - 1 \right] \right\} \mathrm{d}t - \frac{u''(c)}{u'(c)}\{\tilde{c} - c\}\mathrm{d}q. \tag{11.1.10}$$

- Setting $\lambda = 0$ gives the deterministic Keynes-Ramsey rule.

11.1.3 The Keynes-Ramsey Rule: CRRA Utility

- With CRRA utility, inverse intertemporal elasticity is σ.

$$u(c(\tau)) = \frac{c(\tau)^{1-\sigma} - 1}{1-\sigma}, \qquad \sigma > 0. \tag{11.1.11}$$

- The Keynes-Ramsey rule becomes

$$\sigma \frac{\mathrm{d}c}{c} = \left\{ r - \rho + \lambda \left[(1+\beta) \left(\frac{c}{\tilde{c}} \right)^{\sigma} - 1 \right] \right\} \mathrm{d}t + \sigma \left\{ \frac{\tilde{c}}{c} - 1 \right\} \mathrm{d}q. \tag{11.1.12}$$

- The $\mathrm{d}q$ term is the discrete consumption adjustment at jump dates.

11.1.4 Optimal Consumption and Portfolio Choice: Assets

- The household chooses consumption and the risky-asset share $\theta(t)$.
- The risky asset follows a geometric Poisson process; the safe asset grows deterministically.

$$\mathrm{d}v_1(t) = r_1 v_1(t)\mathrm{d}t + \beta v_1(t)\mathrm{d}q(t), \qquad \beta > 0, \tag{11.1.13}$$

$$\mathrm{d}v_2(t) = r_2 v_2(t)\mathrm{d}t, \qquad r_2 \geq 0. \tag{11.1.14}$$

- Wage income is fixed at w and consumption satisfies $c(t) \geq 0$.

11.1.4 Optimal Consumption and Portfolio Choice: Budget Constraint

- The portfolio share $\theta(t)$ determines both the deterministic return and the exposure to jumps.

$$\mathrm{d}a(t) = \{[\theta(t)r_1 + (1-\theta(t))r_2]a(t) + w - c(t)\}\mathrm{d}t + \beta\theta(t)a(t)\mathrm{d}q(t). \tag{11.1.15}$$

- To avoid a trivial portfolio problem, assume

$$r_1 < r_2 < r_1 + \lambda\beta. \tag{11.1.16}$$

- The safe return exceeds the guaranteed risky return, but not the expected risky return.

11.1.4 Optimal Consumption and Portfolio Choice: Bellman Equation

- Let $\tilde{a} \equiv (1+\theta\beta)a$ be wealth after a risky-asset price jump.

$$\rho V(a) = \max_{c(t),\theta(t)} \Big\{ u(c) + [(\theta r_1 + (1-\theta) r_2)a + w - c] V'(a) + \lambda [V(\tilde{a}) - V(a)] \Big\}. \tag{11.1.17}$$

- The two controls generate two first-order conditions.

$$u'(c) = V'(a), \tag{11.1.18}$$

$$V'(a)(r_1 - r_2)a + \lambda V'(\tilde{a})\beta a = 0. \tag{11.1.19}$$

11.1.4 Optimal Consumption and Portfolio Choice: Consumption Jump

- Combining the consumption FOC and portfolio FOC pins down the ratio of post-jump to pre-jump marginal utility.

$$\frac{u'(\tilde{c})}{u'(c)} = \frac{r_2 - r_1}{\lambda\beta}, \qquad a \neq 0. \tag{11.1.20}$$

- With CRRA utility, the consumption jump is explicit:

$$\frac{\tilde{c}}{c} = \left(\frac{\lambda\beta}{r_2 - r_1}\right)^{1/\sigma}. \tag{11.1.21}$$

- By (11.1.16), the right-hand side is larger than one.

11.1.4 Optimal Consumption and Portfolio Choice: DP2

- The CVF for the shadow value under portfolio exposure is

$$\begin{aligned} \mathrm{d}V'(a) =& \{[\theta r_1 + (1-\theta)r_2]a + w - c\}V''(a)\mathrm{d}t \\ &+ \{V'(\tilde{a}) - V'(a)\}\mathrm{d}q(t). \end{aligned} \tag{11.1.22}$$

- Differentiate the maximized Bellman equation by the envelope theorem:

$$\begin{aligned} \rho V'(a) =& \{[\theta r_1 + (1-\theta)r_2]a + w - c\}V''(a) \\ &+ \{\theta r_1 + (1-\theta)r_2\}V'(a) + \lambda\{V'(\tilde{a})[1+\theta\beta] - V'(a)\}. \end{aligned}$$

11.1.4 Optimal Consumption and Portfolio Choice: Costate Dynamics

- Substitute the envelope derivative into the CVF for $V'(a)$.

$$\begin{aligned} \mathrm{d}V'(a) = &\Big\{ \big[\rho - (\theta r_1 + (1-\theta) r_2)\big] V'(a) \\ &\quad - \lambda \big[[1 + \theta\beta] V'(\tilde{a}) - V'(a) \big] \Big\} \mathrm{d}t \\ &+ \{ V'(\tilde{a}) - V'(a) \} \mathrm{d}q(t). \end{aligned}$$

- This is the costate law before inserting the first-order conditions.

11.1.4 Optimal Consumption and Portfolio Choice: Keynes-Ramsey Rule

- Replace $V'(a)$ by $u'(c)$ and use the portfolio FOC to eliminate $\tilde{c}$ from the drift term.

$$\begin{aligned} \mathrm{d}u'(c) = &\Big\{[\rho - (\theta r_1 + (1-\theta) r_2)]u'(c) \\ &\quad - \lambda\big[[1+\theta\beta]u'(\tilde{c}) - u'(c)\big]\Big\}\mathrm{d}t + \{u'(\tilde{c}) - u'(c)\}\mathrm{d}q(t), \end{aligned}$$

$$\frac{u''(c)}{u'(c)}\mathrm{d}c = \left\{\theta r_1 + (1-\theta) r_2 - \rho + \lambda\left[\frac{r_2 - r_1}{\lambda\beta}(1+\theta\beta) - 1\right]\right\}\mathrm{d}t - \frac{u''(c)}{u'(c)}(\tilde{c} - c)\mathrm{d}q(t).$$

11.1.4 Optimal Consumption and Portfolio Choice: CRRA Rule

- For CRRA preferences, the optimal consumption law can be written only in terms of parameters and θ.

$$\frac{\mathrm{d}c(t)}{c(t)} = \frac{1}{\sigma}\left\{r_2 - \lambda\left[1 - \frac{r_2 - r_1}{\lambda\beta}\right] - \rho\right\}\mathrm{d}t + \left\{\left(\frac{\lambda\beta}{r_2 - r_1}\right)^{1/\sigma} - 1\right\}\mathrm{d}q(t).$$

- Consumption grows smoothly between jumps and discretely increases when the risky asset price jumps.

11.1.5 Other Ways to Determine $\tilde{c}$

- Without portfolio choice, the level $\tilde{c}$ after a jump is not pinned down by a portfolio FOC.
- The issue is analogous to choosing the initial consumption level in deterministic Ramsey dynamics.
- In principle, $\tilde{c}$ is determined by a transversality condition.

$$\sigma \frac{\mathrm{d}c}{c} = \left\{ r - \rho + \lambda \left[(1+\beta) \left(\frac{c}{\tilde{c}} \right)^{\sigma} - 1 \right] \right\} \mathrm{d}t + \sigma \left\{ \frac{\tilde{c}}{c} - 1 \right\} \mathrm{d}q.$$

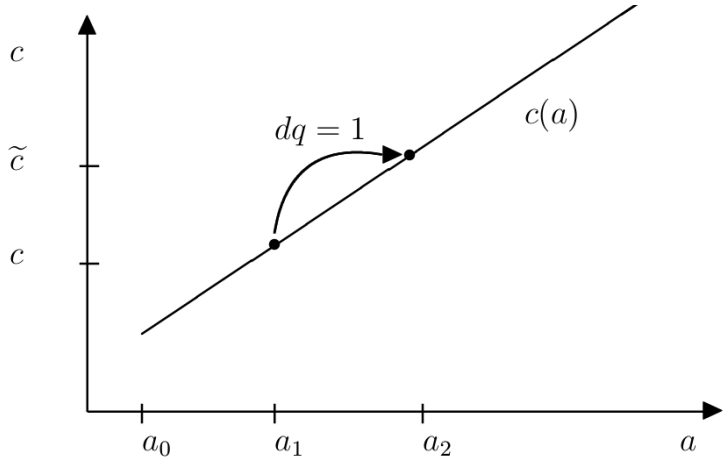


Figure 11.1.1 *Consumption c as a function of wealth a*

11.1.5 Other Ways to Determine $\tilde{c}$: Idiosyncratic Risk

- With one asset and idiosyncratic risk, aggregate variables do not depend on the individual Poisson process.
- If the value function depends only on wealth, the FOC implies a consumption policy function.

$$u'(c) = V'(a) \quad (p = 1), \qquad c = c(a).$$

- After a jump from a_1 to a_2, consumption satisfies

$$\tilde{c} = c(a_2), \qquad a_2 = (1 + \beta)a_1.$$

11.1.5 Other Ways to Determine $\tilde{c}$: Value Functions

- A closed-form value function can sometimes be guessed and verified.
- For the saving-investment problem in 11.1.4, a solution has the form

$$V(a) = \frac{\varphi^{\sigma}\left(a + \frac{w}{r_2}\right)^{1-\sigma} - \rho^{-1}}{1-\sigma},$$

$$c = \varphi\left[a + \frac{w}{r_2}\right], \qquad \theta = \left[\left(\frac{\lambda\beta}{r_2 - r_1}\right)^{1/\sigma} - 1\right]\frac{a + w/r_2}{\beta a}.$$

- Consumption is a constant share of financial plus human wealth.

11.1.5 Other Ways to Determine $\tilde{c}$: Special Cases

- For many standard models, closed-form solutions do not exist for general parameter sets.
- Economists then either solve the model numerically or impose useful parameter restrictions.
- Useful restrictions restore explicit value functions and closed-form policy rules.

general model $\Rightarrow$ numerical policy functions, restricted parameters $\Rightarrow$ closed-form solutions.

11.2 Matching on the Labour Market

- Matching models use value functions for employment and unemployment.
- These value functions can be derived from a household optimization problem.

value functions in matching models $\Longleftarrow$ Bellman equation with labour-income transitions.

- The derivation clarifies special assumptions such as no saving or steady state.

11.2.1 A Household

- Wealth evolves deterministically conditional on labour income z.
- Labour income switches between wage income w and unemployment benefits b.

$$\mathrm{d}a = \{ra + z - c\}\mathrm{d}t.$$

$$\mathrm{d}z = gz\mathrm{d}t - \Delta\mathrm{d}q_w + \Delta\mathrm{d}q_b, \qquad \Delta \equiv w - b.$$

- Job destruction and job creation arrive with state-dependent rates $s(z)$ and $\lambda(z)$.

11.2.1 A Household: State-Dependent Arrival Rates

- An employed worker cannot find a job; an unemployed worker cannot lose one.

	$z = w$	$z = b$
$\lambda(z)$	0	λ
$s(z)$	s	0

- The household maximizes expected lifetime utility with CES/CRRA instantaneous utility.

$$E_t \int_t^\infty e^{-\rho(\tau - t)} u(c(\tau))\, d\tau, \qquad u(c) = \frac{c^{1-\sigma} - 1}{1 - \sigma}, \quad \sigma > 0.$$

11.2.2 The Bellman Equation and Value Functions

- The state variables are wealth a and labour income z.

$$\rho V(a,z) = \max_c \left\{ u(c) + \frac{1}{\mathrm{d}t} E_t \mathrm{d}V(a,z) \right\}.$$

$$\begin{aligned} \mathrm{d}V(a,z) =& \{V_a[ra + z - c] + V_z g z\}\mathrm{d}t \\ &+ [V(a, z - \Delta) - V(a,z)]\mathrm{d}q_w \\ &+ [V(a, z + \Delta) - V(a,z)]\mathrm{d}q_b. \end{aligned}$$

11.2.2 The Bellman Equation and Value Functions: General Form

- Taking expectations with $E_t \mathrm{d}q_w = s(z)\mathrm{d}t$ and $E_t \mathrm{d}q_b = \lambda(z)\mathrm{d}t$ gives

$$\begin{aligned}\rho V(a,z) = \max_c \Big\{ & u(c) + V_a[ra + z - c] + V_z g z \\ & + s(z)[V(a, z-\Delta) - V(a,z)] \\ & + \lambda(z)[V(a, z+\Delta) - V(a,z)] \Big\}. \end{aligned} \tag{11.2.1}$$

- Matching value functions are special cases of this Bellman equation.

11.2.2 The Bellman Equation and Value Functions: Unemployment

- Let $U \equiv V(a, b)$ and $W \equiv V(a, w)$.
- Since $s(b) = 0$ and $\lambda(b) = \lambda$, the unemployment Bellman equation is

$$\rho U = \max_c \{u(c) + U_a[ra + b - c] + U_z gb + \lambda[W - U]\}.$$

- Under optimal controls,

$$\rho U = u(c) + U_a[ra + b - c] + U_z gb + \lambda[W - U].$$

11.2.2 The Bellman Equation and Value Functions: Pissarides Form

- If households cannot save, $c = ra + b$.
- If there is no technological progress, $g = 0$.

$$\rho U = u(ra + b) + \lambda[W - U].$$

- With risk neutrality $u(c) = c$, no capital income $a = 0$, and $r = \rho$:

$$rU = b + \lambda[W - U].$$

- This is the standard value equation for unemployment in the matching literature.

11.3 Intertemporal Utility Maximization under Brownian Motion

- The problem now uses Brownian rather than Poisson uncertainty.
- Wealth follows a diffusion with drift $ra + w - pc$ and volatility proportional to wealth.

$$\mathrm{d}a = \{ra + w - pc\}\mathrm{d}t + \beta a \mathrm{d}z.$$

$$U(t) = E_t \int_t^\infty e^{-\rho(\tau - t)} u(c(\tau))\, \mathrm{d}\tau, \qquad V(a) = \max_{\{c(\tau)\}} U(t).$$

11.3.2 Solving by Dynamic Programming: Bellman Equation

- Ito's Lemma gives the differential of the value function.

$$\mathrm{d}V(a) = \left\{ V'(a)[ra + w - pc] + \frac{1}{2}V''(a)\beta^2 a^2 \right\} \mathrm{d}t + V'(a)\beta a \mathrm{d}z.$$

- Taking expectations gives the Bellman equation:

$$\rho V(a) = \max_{c(t)} \left\{ u(c(t)) + V'(a)[ra + w - pc] + \frac{1}{2}V''(a)\beta^2 a^2 \right\}.$$

$$u'(c(t)) = V'(a)p(t). \tag{11.3.1}$$

11.3.2 Solving by Dynamic Programming: DP2

- Differentiate the maximized Bellman equation with respect to wealth.

$$\rho V' = V''[ra + w - pc] + V'r + \frac{1}{2}V'''\beta^2 a^2 + V''\beta^2 a,$$
$$(\rho - r)V' = V''[ra + w - pc] + \frac{1}{2}V'''\beta^2 a^2 + V''\beta^2 a. \tag{11.3.2}$$

- The third derivative appears because the Bellman equation contains V''.

11.3.2 Solving by Dynamic Programming: Costate Dynamics

- Apply Ito's Lemma to the shadow price $V'(a)$.

$$\begin{aligned} \mathrm{d}V' &= V''\mathrm{d}a + \frac{1}{2}V'''\beta^2 a^2 \mathrm{d}t \\ &= V''[ra + w - pc]\mathrm{d}t + \frac{1}{2}V'''\beta^2 a^2 \mathrm{d}t + V''\beta a \mathrm{d}z. \end{aligned}$$

- Substituting from (11.3.2) yields

$$\mathrm{d}V' = \{(\rho - r)V' - V''\beta^2 a\}\mathrm{d}t + V''\beta a \mathrm{d}z. \tag{11.3.3}$$

11.3.2 Solving by Dynamic Programming: DP3

- Use the FOC and allow the consumption price p to vary independently of V'.

$$\mathrm{d}u'(c) = p\mathrm{d}V' + V'\mathrm{d}p.$$

$$\begin{aligned}\mathrm{d}u'(c) = &\left\{(\rho - r)u'(c) - u''(c)c'(a)\beta^2 a\right\}\mathrm{d}t \\ &+ u''(c)c'(a)\beta a\mathrm{d}z + u'(c)\frac{\mathrm{d}p}{p}.\end{aligned} \tag{11.3.4}$$

- The diffusion term affects both the stochastic term and the deterministic drift.

11.3.3 The Keynes-Ramsey Rule

- Apply the inverse marginal-utility transformation to (11.3.4).

$$\frac{u''}{u'}\mathrm{d}c = \left\{ r - \rho + \frac{u''(c)}{u'(c)} c_a \beta^2 a + \frac{1}{2}\frac{u'''(c)}{u'(c)}[c_a \beta a]^2 \right\} \mathrm{d}t - \frac{u''(c)}{u'(c)} c_a \beta a \mathrm{d}z - \frac{\mathrm{d}p}{p}. \tag{11.3.5}$$

- Here c_a denotes the derivative of the consumption policy with respect to wealth.

11.3.3 The Keynes-Ramsey Rule: CRRA

- For CRRA utility, substitute u', u'', and u''' into the general rule.

$$\sigma \frac{\mathrm{d}c}{c} = \left\{ r - \rho - \sigma \frac{c_a}{c} \beta^2 a + \frac{1}{2} \sigma(\sigma + 1) \left[\frac{c_a}{c} \beta a \right]^2 \right\} \mathrm{d}t + \sigma \frac{c_a}{c} \beta a \mathrm{d}z - \frac{\mathrm{d}p}{p}. \tag{11.3.6}$$

- Brownian uncertainty affects the drift through second-order terms.

11.4 Capital Asset Pricing

- The capital asset pricing application follows the same Bellman-FOC logic.
- Wealth consists of a portfolio of N assets.

$$a = \sum_{i=1}^{N} p_i n_i.$$

- Asset prices follow geometric Brownian motion:

$$\frac{\mathrm{d}p_i}{p_i} = \alpha_i \mathrm{d}t + \sigma_i \mathrm{d}z_i. \tag{11.4.1}$$

11.4.1 The setup: Portfolio Adjustment

- Households allocate shares χ_i of savings across assets.

$$\mathrm{d}n_i = \frac{1}{p_i}\chi_i \left\{ \sum_{i=1}^{N} \pi_i n_i + w - c \right\} \mathrm{d}t. \tag{11.4.2}$$

- Wealth in asset i changes according to

$$\mathrm{d}(p_i n_i) = p_i \mathrm{d}n_i + n_i \mathrm{d}p_i = \chi_i \left\{ \sum_{i=1}^{N} \pi_i n_i + w - c \right\} \mathrm{d}t + n_i \mathrm{d}p_i.$$

11.4.1 The setup: Budget Constraint

- Let $a_i = n_i p_i$ and define the portfolio share $\theta_i \equiv a_i / a$.

$$\theta_N = 1 - \sum_{i=1}^{N-1} \theta_i. \tag{11.4.3}$$

$$r_i \equiv \frac{\pi_i}{p_i} + \alpha_i, \qquad r \equiv \sum_{i=1}^{N} \theta_i r_i. \tag{11.4.4}$$

- The household budget constraint is

$$\mathrm{d}a = \{ra + w - c\}\mathrm{d}t + a \sum_{i=1}^{N} \theta_i \sigma_i \mathrm{d}z_i. \tag{11.4.5}$$

11.4.2 Optimal Behaviour: Expected Change in Value

- For one state variable, Ito's Lemma gives

$$\frac{1}{\mathrm{d}t}E_t\mathrm{d}V(a) = \frac{1}{\mathrm{d}t}E_t\left(V'(a)\mathrm{d}a + \frac{1}{2}V''(a)[\mathrm{d}a]^2\right). \tag{11.4.6}$$

- With covariances $\sigma_{ij} \equiv \sigma_i\sigma_j\rho_{ij}$,

$$\begin{aligned}[\mathrm{d}a]^2 &= a^2[\theta_1\sigma_1\mathrm{d}z_1 + \cdots + \theta_N\sigma_N\mathrm{d}z_N]^2 \\ &= a^2\sum_{j=1}^{N}\sum_{i=1}^{N}\theta_i\theta_j\sigma_{ij}\mathrm{d}t.\end{aligned} \tag{11.4.7}$$

11.4.2 Optimal Behaviour: Portfolio Variance

- Use $\theta_N = 1 - \sum_{i=1}^{N-1} \theta_i$ to express the variance in terms of $N-1$ independent shares.

$$(\mathrm{d}a)^2 = a^2 \Bigg\{ \sum_{j=1}^{N-1} \sum_{i=1}^{N-1} \theta_j \theta_i (\sigma_{ij} - 2\sigma_{iN}) + 2 \sum_{i=1}^{N-1} \theta_i \sigma_{iN} + \left(1 - \sum_{i=1}^{N-1} \theta_i \right)^2 \sigma_N^2 \Bigg\} \mathrm{d}t. \tag{11.4.8}$$

- The market return can also be rewritten relative to asset N:

$$r = r_N + \sum_{i=1}^{N-1} \theta_i (r_i - r_N). \tag{11.4.9}$$

11.4.2 Optimal Behaviour: Bellman Equation

- The Bellman equation is obtained by substituting (11.4.8) and (11.4.9).

$$\rho V(a) = \max_{c(t),\theta_i(t)} \left\{ u(c) + V'(a) \left([r_N + \sum_{i=1}^{N-1} \theta_i (r_i - r_N)]a + w - c \right) + \frac{1}{2} V''(a)(\mathrm{d}a)^2/\mathrm{d}t \right\}.$$

- The consumption FOC is the standard marginal-utility condition:

$$u'(c) = V'(a).$$

11.4.2 Optimal Behaviour: Portfolio FOC

- Differentiating the variance term with respect to θ_i gives

$$\frac{\mathrm{d}\{\cdot\}}{\mathrm{d}\theta_i} = 2\sum_{j=1}^{N} \theta_j[\sigma_{ij} - \sigma_{iN}]. \tag{11.4.10}$$

- The portfolio first-order condition is

$$V'a(r_i - r_N) + \frac{1}{2}V''2a^2\sum_{j=1}^{N} \theta_j[\sigma_{ij} - \sigma_{iN}] = 0,$$

$$r_i - r_N = -\frac{V''}{V'}a\sum_{j=1}^{N} \theta_j[\sigma_{ij} - \sigma_{iN}]. \tag{11.4.11}$$

11.4.3 Capital Asset Pricing

- Assume asset N is riskless: $\sigma_N = 0$ and $\sigma_{Nj} = 0$ for all j.
- Define

$$\Omega \equiv -a\frac{V''}{V'}, \qquad \sigma_{iM} \equiv \sum_{j=1}^{N} \theta_j \sigma_{ij}, \qquad \sigma_M^2 \equiv \sum_{i=1}^{N} \theta_i \sigma_{iM}, \qquad r \equiv \sum_{i=1}^{N} \theta_i r_i.$$

- The portfolio FOC becomes

$$r_i - r_N = \Omega \sigma_{iM}. \tag{11.4.12}$$

11.4.3 Capital Asset Pricing: CAP Equation

- Multiply (11.4.12) by θ_i and sum across assets:

$$\sum_{i=1}^{N} \theta_i (r_i - r_N) = \Omega \sum_{i=1}^{N} \theta_i \sigma_{iM} \quad \Longrightarrow \quad r - r_N = \Omega \sigma_M^2.$$

- Dividing by (11.4.12) yields the capital asset pricing equation:

$$r_i - r_N = \frac{\sigma_{iM}}{\sigma_M^2}(r - r_N).$$

- The ratio σ_{iM}/σ_M^2 is the usual beta factor.

11.5 Natural Volatility II

- Natural volatility models explain growth and fluctuations through occasional technology jumps.
- Poisson processes are useful because economically relevant technology changes are rare.

$$\text{growth cycles} \quad \Longleftarrow \quad \text{positive TFP jumps.}$$

- The section gives a simplified stochastic continuous-time RBC setup.

11.5.1 An Real Business Cycle Model: Technologies

- Final output is produced with Cobb-Douglas technology.

$$Y = K^{\alpha}(AL)^{1-\alpha}. \tag{11.5.1}$$

- Labour productivity follows a geometric Poisson process:

$$\frac{\mathrm{d}A}{A} = g\mathrm{d}t + \gamma \mathrm{d}q. \tag{11.5.2}$$

- The expected growth rate of A is $g + \lambda\gamma$.

11.5.1 An Real Business Cycle Model: Capital Accumulation

- The final good is used for consumption and investment, $Y = C + I$.
- With Y as numeraire, all final-good prices equal one.

$$\mathrm{d}K = (Y - C - \delta K)\mathrm{d}t. \tag{11.5.3}$$

- Firms maximize instantaneous profits and pay factor prices equal to marginal products.

$$r = \frac{\partial Y}{\partial K} - \delta.$$

11.5.1 An Real Business Cycle Model: Households

- Households maximize expected lifetime utility and hold wealth in firm shares k.

$$U(t) = E_t \int_t^\infty e^{-\rho(\tau - t)} u(c(\tau)) \mathrm{d}\tau, \qquad u(c) = \frac{c^{1-\sigma} - 1}{1 - \sigma}. \tag{11.5.4}$$

- Wealth measured in physical capital evolves deterministically conditional on prices:

$$\mathrm{d}k = (rk + w - c)\mathrm{d}t. \tag{11.5.5}$$

11.5.1 An Real Business Cycle Model: Optimal Behaviour

- Households take both k and A as state variables.
- Although individual wealth has no direct jump term, consumption jumps when TFP jumps.

$$\sigma \frac{\mathrm{d}c}{c} = \left\{ r - \rho + \lambda \left[\left(\frac{c}{\tilde{c}} \right)^{\sigma} - 1 \right] \right\} \mathrm{d}t + \left\{ \frac{\tilde{c}}{c} - 1 \right\} \mathrm{d}q. \tag{11.5.6}$$

- A technology jump changes wages and interest rates, so optimal consumption adjusts discretely.

11.5.1 An Real Business Cycle Model: General Equilibrium

- The aggregate economy is described by labour productivity, capital, and consumption.

$$\frac{\mathrm{d}A}{A} = g\mathrm{d}t + \gamma \mathrm{d}q, \qquad \mathrm{d}K = \left(K^{\alpha}(AL)^{1-\alpha} - C - \delta K\right)\mathrm{d}t.$$

$$\sigma \frac{\mathrm{d}C}{C} = \left\{ \alpha \left[\frac{AL}{K}\right]^{1-\alpha} - \delta - \rho + \lambda \left[\left(\frac{C}{\tilde{C}}\right)^{\sigma} - 1\right] \right\} \mathrm{d}t + \left\{ \frac{\tilde{C}}{C} - 1 \right\} \mathrm{d}q.$$

- Relative to deterministic models, the key new objects are $\mathrm{d}q$ and $\tilde{C}$.

11.5.1 An Real Business Cycle Model: Parameter Restriction

- A constant saving rate $s = 1 - \sigma^{-1}$ is optimal for a special parameter set.

$$C = (1 - s)Y, \qquad s = 1 - \sigma^{-1},$$

$$\sigma = \frac{\rho + \delta - \lambda[(1 + \gamma)^{1-\alpha} - 1]}{\alpha\delta - (1 - \alpha)g}.$$

- The meaningful case requires $\sigma > 1$, so the intertemporal elasticity σ^{-1} is below one.

11.5.1 An Real Business Cycle Model: Detrended Capital

- Use capital per effective worker to remove the trend from capital.

$$\hat{K} = \frac{K}{A}.$$

- Applying the appropriate CVF gives

$$\mathrm{d}\hat{K} = \left\{\beta_1 \hat{K}^{\alpha} L^{1-\alpha} - \beta_2 \hat{K}\right\} \mathrm{d}t - \beta_3 \hat{K} \mathrm{d}q, \tag{11.5.7}$$

$$0 < \beta_3 < 1.$$

- A technology jump raises A instantly and reduces $\hat{K} = K/A$.

11.5.1 An Real Business Cycle Model: Cyclical Growth

- If $\mathrm{d}q = 0$, $\hat{K}$ moves smoothly toward the temporary steady state.

$$\mathrm{d}\hat{K} = 0, \quad \mathrm{d}q = 0 \quad \Longrightarrow \quad \hat{K}^* = \left(\frac{\beta_1 L^{1-\alpha}}{\beta_2} \right)^{1/(1-\alpha)}.$$

- A technology jump lowers $\hat{K}$ on impact by raising the denominator A.
- GDP jumps upward and then approaches a new higher upper limit.

$$Y_q \longrightarrow Y_q^*, \qquad q \mapsto q+1: \quad Y_q^* \longrightarrow Y_{q+1}^*.$$

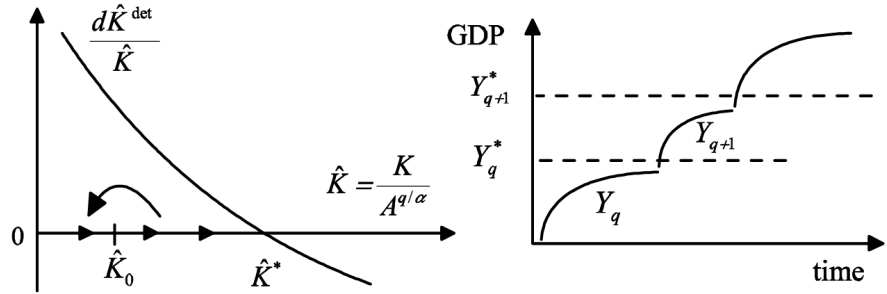


Figure 11.5.1 *Cyclical growth*

11.5.2 A Natural Volatility Model

- The RBC setup can be extended to endogenous technology jumps.
- Resources invested in R&D determine the arrival rate of technology jumps.

$$\lambda = \lambda(R).$$

- The resource constraint becomes

$$\mathrm{d}K = (Y - C - R - \delta K)\mathrm{d}t.$$

- Larger R&D investment raises the frequency of jumps and shortens business cycles.

11.5.3 A Numerical Approach: Stationary Variables

- For numerical work, write the model in stationary variables.

$$\hat{K} = \frac{K}{A}, \qquad \hat{C} = \frac{C}{A}.$$

- For exogenous arrival rates, the transformed dynamics are

$$\sigma \frac{\mathrm{d}\hat{C}}{\hat{C}} = \left\{ \alpha \left(\frac{L}{\hat{K}} \right)^{1-\alpha} - \delta - \rho - \sigma g + \lambda \left[\left(\frac{\hat{C}}{(1+\gamma)\tilde{\hat{C}}} \right)^{\sigma} - 1 \right] \right\} \mathrm{d}t + \sigma \left\{ \frac{\tilde{\hat{C}}}{\hat{C}} - 1 \right\} \mathrm{d}q, \tag{11.5.8}$$

$$\mathrm{d}\hat{K} = \left\{ \hat{Y} - (\delta + g)\hat{K} - \hat{C} \right\} \mathrm{d}t - \frac{\gamma}{1+\gamma} \hat{K} \mathrm{d}q. \tag{11.5.9}$$

11.5.3 A Numerical Approach: Policy Function

- The solution is a policy function $\hat{C}(\hat{K})$, analogous to a saddle path.
- The post-jump consumption term is evaluated at the post-jump detrended capital stock.

$$\tilde{\hat{C}} = \hat{C}\left(\frac{1}{1+\gamma}\hat{K}\right).$$

- Since $\gamma > 0$, a jump reduces technology-transformed capital $\hat{K}$.
- The numerical trick: start with an initial path $\hat{C}(\hat{K})$ and treat it as exogenous inside (11.5.8).

$$\text{initial path } \hat{C}(\hat{K}) \quad \Rightarrow \quad \text{ordinary differential equation system.}$$